

AI Search Strategy Questions (Worksheet)

Generated from Knowledge Graph Analysis

This document contains problem instances for various AI search problems. Please analyze each instance and determine the most appropriate solving strategy. You can type your answers directly into the text boxes provided.

1. N-Queens

Instance 1:

Board Size: 6×6, Pre-placed Queens: 1

.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	Q	.

Question: For the N-Queens problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Your Answer:

Instance 2:

Board Size: 5×5, Pre-placed Queens: 2

.	.	.	.	.
Q	.	.	.	.
.	.	.	Q	.
.	.	.	.	.
.	.	.	.	.

Question: For the N-Queens problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Your Answer:

Instance 3:

Board Size: 5×5, Pre-placed Queens: 0

•	•	•	•	•
•	•	•	•	•
•	•	•	•	•
•	•	•	•	•
•	•	•	•	•

Question: For the N-Queens problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Your Answer:

2. Tower of Hanoi

Instance 1:

Number of Disks: 3

Initial Configuration:

Peg A: [3, 2, 1]

Peg B: []

Peg C: []

Goal: All disks on Peg C

Question: For the Tower of Hanoi problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Your Answer:

Instance 2:

Number of Disks: 3

Initial Configuration:

Peg A: [3, 1]

Peg B: []

Peg C: [2]

Goal: All disks on Peg C

Question: For the Tower of Hanoi problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Your Answer:

Instance 3:

Number of Disks: 4

Initial Configuration:

Peg A: [4, 3, 2, 1]

Peg B: []

Peg C: []

Goal: All disks on Peg C

Question: For the Tower of Hanoi problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Your Answer:

3. Graph Coloring

Instance 1:

Vertices: 5, **Colors Available:** 3, **Edges:** 6

Graph Edges:

(0, 1), (0, 2), (1, 2), (1, 3), (2, 4), (3, 4)

Question: For the Graph Coloring problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Your Answer:

Instance 2:

Vertices: 7, **Colors Available:** 4, **Edges:** 8

Graph Edges:

(0, 1), (0, 4), (1, 4), (2, 3), (2, 5), (2, 6), (3, 4), (3, 6)

Question: For the Graph Coloring problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Your Answer:

Instance 3:

Vertices: 7, **Colors Available:** 3, **Edges:** 9

Graph Edges:

(0, 4), (0, 5), (1, 4), (1, 6), (2, 6), (3, 4), (3, 6), (4, 6), (5, 6)

Question: For the Graph Coloring problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Your Answer:

4. Knight's Tour

Instance 1:

Board Size: 5×5, **Starting Position:** (2, 0), **Visited Squares:** 7

.	.	.	6	.
.	7	.	3	.
1	.	5	.	.
.	.	2	.	4
.	.	.	.	.

Question: For the Knight's Tour problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Your Answer:

Instance 2:

Board Size: 5×5, **Starting Position:** (4, 0), **Visited Squares:** 6

.	.	.	.	.
.	.	4	.	.
3	.	.	.	5
.	.	2	.	.
1	.	.	6	.

Question: For the Knight's Tour problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Your Answer:

Instance 3:

Board Size: 5×5, **Starting Position:** (1, 0), **Visited Squares:** 5

.	.	.	.	.
1	.	3	.	.
.	5	.	.	.
.	2	.	4	.
.	.	.	.	.

Question: For the Knight's Tour problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Your Answer:

5. 8-Puzzle

Instance 1:

Initial State (Goal: 1-2-3-4-5-6-7-8-blank)

5	4	2
7	1	3
8		6

Question: For the 8-Puzzle problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Your Answer:

Instance 2:

Initial State (Goal: 1-2-3-4-5-6-7-8-blank)

4	3	6
5	7	1
2		8

Question: For the 8-Puzzle problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Your Answer:

Instance 3:

Initial State (Goal: 1-2-3-4-5-6-7-8-blank)

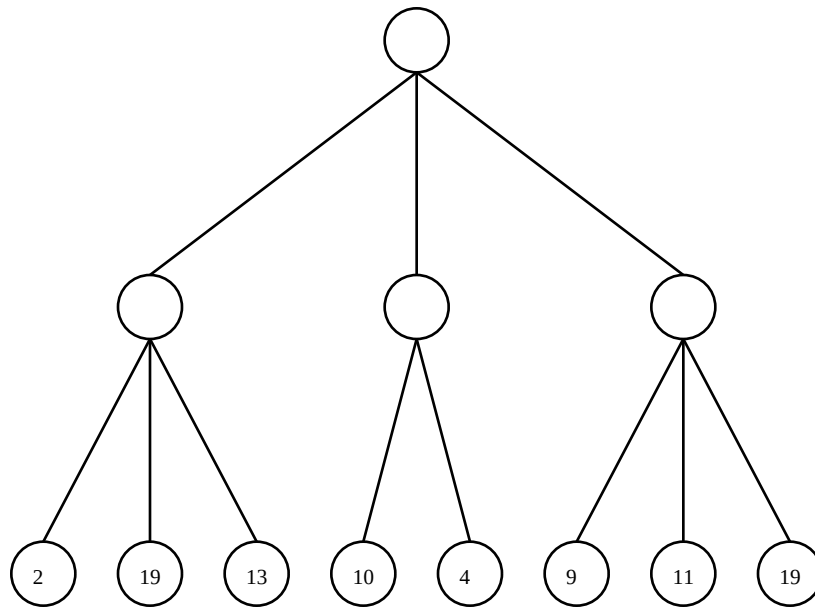
4	1	5
7	3	2
8		6

Question: For the 8-Puzzle problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Your Answer:

6. MinMax

Instance 1:



Question: For the given MinMax tree, what is the value at the root and how many leaf nodes are visited when applying MinMax with Alpha-Beta pruning?

Your Answer:

Root Value:

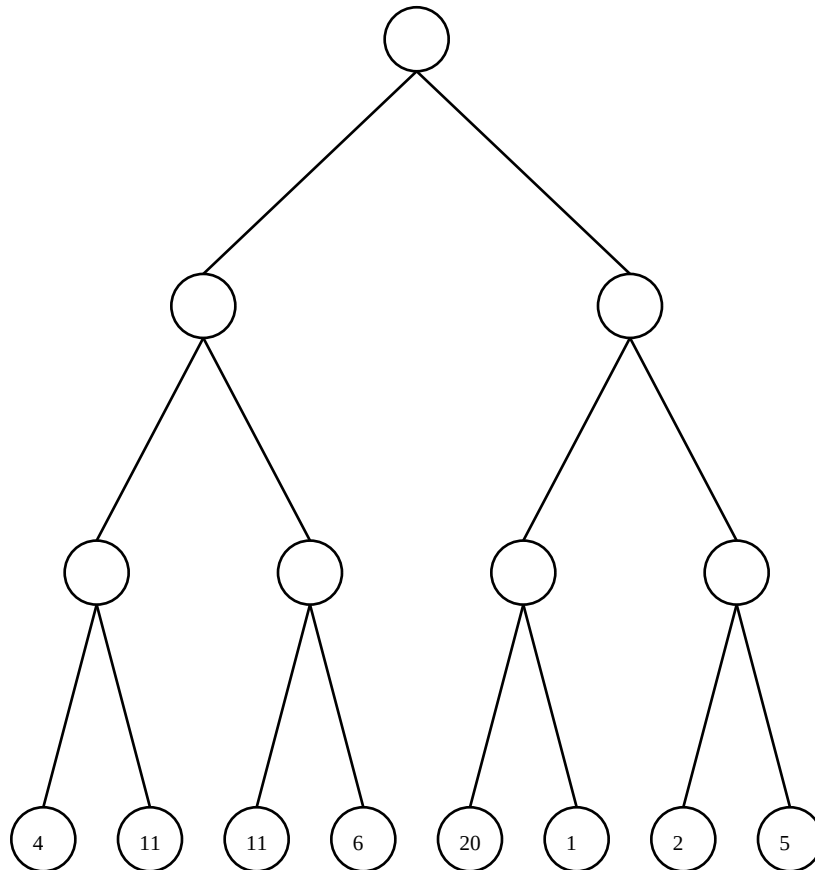
Leaves Visited:

Status:

Check Answer

Note: This interactive check requires a PDF viewer with JavaScript support (e.g., Adobe Acrobat Reader, Foxit Reader, or Master PDF Editor). It may not work in browser-based viewers.

Instance 2:



Question: For the given MinMax tree, what is the value at the root and how many leaf nodes are visited when applying MinMax with Alpha-Beta pruning?

Your Answer:

Root Value:

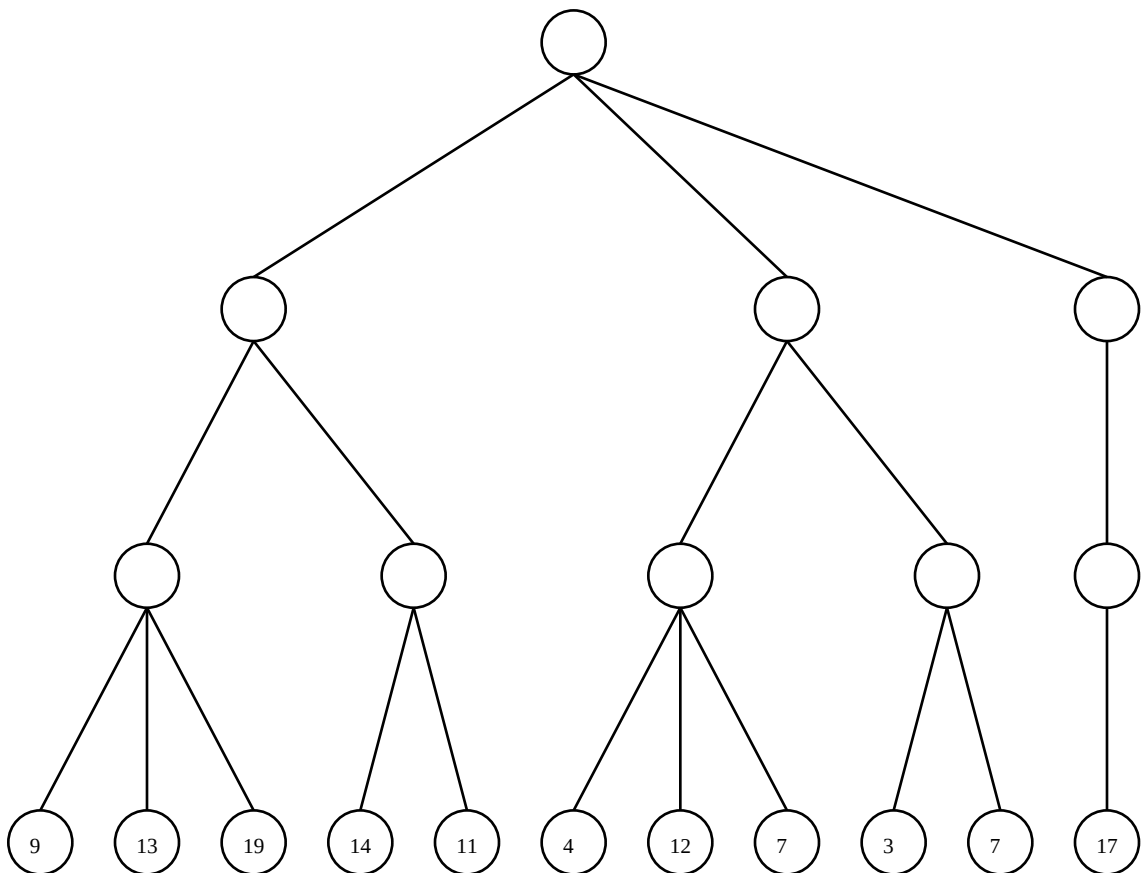
Leaves Visited:

Status:

Check Answer

Note: This interactive check requires a PDF viewer with JavaScript support (e.g., Adobe Acrobat Reader, Foxit Reader, or Master PDF Editor). It may not work in browser-based viewers.

Instance 3:



Question: For the given MinMax tree, what is the value at the root and how many leaf nodes are visited when applying MinMax with Alpha-Beta pruning?

Your Answer:

Root Value:

Leaves Visited:

Status:

Check Answer

Note: This interactive check requires a PDF viewer with JavaScript support (e.g., Adobe Acrobat Reader, Foxit Reader, or Master PDF Editor). It may not work in browser-based viewers.